

Shiv Shankar Singh

+91-7068900427 | singhshiv0427@gmail.com | Portfolio | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Full-stack developer (React, Node.js, TypeScript) building **AI backends from scratch**. Built and deployed DocMind, a RAG app with pgvector search, SSE streaming and Zod-validated LLM output, written **without LangChain or an LLM SDK** to understand how each piece works. Self-taught since graduating in 2024, with a **400+ day GitHub streak** and **100+ days** of structured DSA practice.

PROJECTS

DocMind: RAG Document Chat | Live App | API | GitHub 2026

TypeScript, Express 5, PostgreSQL, pgvector, Drizzle ORM, Gemini API, Zod, SSE, React 19

- Built the **RAG pipeline** from scratch, PDFs are chunked, embedded with Gemini and stored in Postgres with **pgvector**, and each question is answered from the closest chunks with sources shown under every reply.
- Streamed answers word by word over **Server-Sent Events** by parsing Gemini's raw byte stream, with **conversation memory** in Postgres so follow-ups and page refreshes keep the same chat.
- Generated quizzes with Gemini structured output **validated by Zod** (retry once, then fail clean), and shuffled options in code after measuring the model put the right answer in one slot **36% of the time**.
- Layered as **routes, services and repositories** so only repositories touch SQL and services never touch req/res.

TaskForge: REST API & Task Manager | Live App | API Docs | GitHub 2026

Node.js, Express 5, MongoDB, Mongoose, JWT, Zod, Swagger, GitHub Actions

- Built a secure REST API (Express 5, MVC) with **JWT auth**, bcrypt hashing and Zod validation on bodies and query params, with every MongoDB query **scoped to the user** from the token so no one sees someone else's tasks.
- Task CRUD with server-side filtering, sorting, pagination, Swagger docs and a 12-check API test script.
- Hardened with Helmet, proxy-aware **IP rate limiting** (100 req/15 min) and CORS locked to the frontend, with GitHub Actions CI and a Render blueprint deploy with health checks.

Cinegraph: AI Recommendation Engine | Live App | GitHub 2026

Gemini API, Cloudflare Workers, Firebase, TMDB API, React 19, Redux Toolkit

- Built a **serverless LLM proxy** on Cloudflare Workers so the **Gemini key never reaches the browser**, with a CORS allow-list and input caps (300-char profile, max 5 history turns) to bound cost and abuse.
- Prompted Gemini for strict JSON and **checked every title against TMDB** so fake titles never reach the UI.
- Personalized picks with a **taste profile** from user ratings (genre scores, avoided genres, favorite decade) in the prompt, and follow-ups like "more like the third one" run as real multi-turn chat.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL

Backend: Node.js, Express.js, REST APIs, SSE, JWT, bcrypt, Zod, Swagger/OpenAPI, Helmet, Rate Limiting

AI / LLM: Gemini API (generation, streaming, embeddings), RAG, Vector Search, Structured Output

Databases: PostgreSQL, pgvector, Drizzle ORM, MongoDB, Mongoose, Firebase Firestore

Frontend: React.js (v19), Redux Toolkit, Tailwind CSS, Vite

DevOps & Tools: Git, GitHub Actions, Render, Vercel, Netlify, Cloudflare Workers, Neon

EDUCATION

Lovely Professional University

Punjab, India

B.Tech in Computer Science & Engineering, CGPA: 7.01

Oct 2020 – Jul 2024

Tulsi Vidya Niketan

Varanasi, India

Higher Secondary (CBSE Board), 83%

Mar 2018 – Jul 2020

CERTIFICATIONS

Web Development Bootcamp, Udemy (Jul 2023) | **Server-side JavaScript with Node.js**, Coursera (May 2023)

DSA Self Paced, GeeksforGeeks (Jul 2022) | **Java Programming**, Coding Ninjas (May 2022)